Magnesium

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Identity Sheet

Property	Value
Name	Magnesium
Other names	Mg, Mg^{2+}
Chemical formula	Mg^{2+} (divalent cation)
Atomic weight	24.31 g/mol
CAS number	7439-95-4
PubChem CID	5462224
Appearance	Colourless in solution

Structural formula:

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Mg(2+) (divalent cation, hydrated in solution)
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Magnesium is the second most abundant intracellular cation and the fourth most abundant cation in the body. It is an essential cofactor for >300 enzymatic reactions, including all ATP-dependent processes. Only ~1% of total body Mg is in ECF; ~60% is in bone, ~39% in soft tissue. Urinary Mg measurement helps differentiate renal from extrarenal causes of hypomagnesaemia.

Molecules Not to Be Confused With

Molecule	Formula	MW (g/mol)	Key Difference
Calcium (Ca^{2+})	Ca^{2+}	40.08	Larger divalent cation; different metabolic roles; often co-measured
Potassium (K^{+})	K^{+}	39.10	Monovalent; primary intracellular cation; different renal handling

Medical Information

Origin

Endogenous Magnesium is not synthesised. Total body Mg: ~24 g (~1000 mmol) in adults. Bone stores act as a reservoir. Released during bone resorption.

Exogenous Dietary sources: green leafy vegetables (chlorophyll contains Mg), nuts, seeds, whole grains, legumes, dark chocolate, mineral water. Typical intake: 300-400 mg/day. Intestinal absorption: 30-50% (small intestine, primarily jejunum and ileum via TRPM6 channels). Mg supplements and antacids are additional sources.

Primary & Secondary Biological Roles

Primary role: - **Enzymatic cofactor:** Required for all kinases (ATP-Mg as true substrate), DNA/RNA polymerases, and >300 other enzymes.

Secondary roles: - **Neuromuscular function:** Modulates calcium channels and NMDA receptors; deficiency causes hyperexcitability, tetany, and seizures. - **Bone structure:** ~60% of body Mg is in bone mineral. - **Cardiac rhythm:** Mg stabilises cardiac cell membranes; deficiency promotes arrhythmias. - **Glucose metabolism:** Insulin signalling requires Mg.

Catabolism and Elimination Pathway

- Mg freely filtered at glomerulus (~2400 mg/day, ~80% of plasma Mg is filterable).
- **~95-97%** reabsorbed: 15% proximal tubule, 70% thick ascending loop of Henle (paracellular via claudin-16/19), 10% distal convoluted tubule (TRPM6, fine-tuning).
- 3-5% excreted in urine (~100 mg/day).
- Minor losses: GI, sweat.

Expression in Humans

Normal Levels

Compartment	Reference Range
Serum Mg	1.7-2.2 mg/dL (0.70-0.95 mmol/L)
Urinary Mg (24-h)	73-122 mg/day (3-5 mmol/day)
Urinary Mg (spot)	Variable; FEMg <4% normally

Factors Influencing Levels **Increased urinary Mg:** - Diuretics (loop » thiazide) - Alcohol excess - Diabetic ketoacidosis (osmotic diuresis) - Cisplatin, aminoglycosides, amphotericin B (nephrotoxic) - Gitelman syndrome, Bartter syndrome - Hypercalcaemia (Ca competes for reabsorption) - IV Mg administration

Decreased urinary Mg: - Low Mg intake - GI Mg losses (diarrhoea, malabsorption) - Appropriate renal conservation in extrarenal hypomagnesaemia (FEMg <2%) - Familial hypomagnesaemia with hypocalciuria

Associated Pathologies

Condition	Mg Pattern	Key Symptoms
Renal Mg wasting	High urinary Mg (FEMg >4%), low serum Mg	Tetany, arrhythmias, seizures
GI Mg loss	Low urinary Mg (FEMg <2%), low serum Mg	Diarrhoea, malabsorption, cramps
Alcoholism	High urinary Mg + poor intake	Tremor, seizures, Wernicke
Drug-induced (PPI, cisplatin)	Variable urinary Mg	Muscle weakness, cramps

Presence in Urine

Should it be normally present? Yes – urinary Mg excretion is the primary regulatory mechanism. Normal: 3-5 mmol/day.

Normal urinary levels: 73-122 mg/day (3-5 mmol/day).

Form in urine: Mg^{2+} ion, hydrated. May complex with phosphate (struvite: $MgNH_4PO_4$) in infected alkaline urine.

Pathological significance:

Urinary Mg	Possible Causes	Prevalence
High (>5 mmol/day)	Renal Mg wasting, drugs, DKA	Common with diuretics
Low (<3 mmol/day)	Extrarenal loss, low intake, renal conservation	Common

Solubility: $MgCl_2$ and $MgSO_4$ are very soluble. $MgNH_4PO_4$ (struvite) is poorly soluble in alkaline conditions and can form stones.

Detection in Urine

Available Clinical Assays

1. **Colorimetric (xylidyl blue / calmagite):**
   - **Principle:** Mg^{2+} reacts with calmagite or xylidyl blue in alkaline pH -> blue-violet complex.
   - **Detection:** Absorbance at **520-550 nm**.
   - **LOD:** ~0.1 mg/dL (~0.04 mmol/L).
   - **Advantages:** Automated, standard on clinical platforms.
2. **Atomic absorption spectroscopy (AAS, reference):**
   - **Principle:** Mg atoms in flame absorb **285.2 nm** light.
   - **Detection:** Absorption at **285.2 nm**.
   - **LOD:** ~0.001 mg/dL.
   - **Advantages:** Highly specific; reference method.
3. **Ion-selective electrode:**
   - **Principle:** Mg^{2+}-selective ionophore membrane (ETH 7025 or similar).
   - **Detection:** Potentiometric.
   - **LOD:** ~0.1 mmol/L. Less common than for Na/K.

Optimal Urine Type for Measurement

- **24-hour urine** for total excretion assessment.
- Spot urine for FEMg calculation (with serum Mg and creatinine).
- Acidify with HCl to prevent Mg salt precipitation.
- Refrigerate.

Actual Gold Standard

Flame AAS at 285.2 nm is the reference method. Colorimetric methods (calmagite/xylidyl blue) on automated platforms are the standard clinical assay. CV: 3-5%.

Interferences in Measurement

Interference	Effect	Mechanism
Calcium (high)	Positive bias (colorimetric)	Ca also reacts with dyes; EGTA chelation masks Ca
Haemolysis	Positive bias	Intracellular Mg release
EDTA contamination	Negative bias	Chelates Mg

Interference	Effect	Mechanism
Lipaemia	Positive bias	Light scattering

Research Detection Methods

Spectroscopy Detection (UV-Vis / NIR)

- **AAS at 285.2 nm:** Reference method. LOD: ~0.001 mg/dL.
- **ICP-OES at 279.6 nm or 285.2 nm:** Multi-element. LOD: ~0.0001 mg/dL.
- **Colorimetric (calmagite, 520 nm):** See clinical assays.
- **NIR:** No useful Mg^{2+} absorption.

Fluorescence Detection

- **Mag-Fura-2 (furaptra):** Ratiometric Mg probe. Ex 340/380 nm, Em 510 nm. Kd ~1.5 mM. LOD: ~0.1 mM.
- **Mag-Indo-1:** Ex 349 nm, Em 390/480 nm. LOD: ~0.1 mM.
- **Mag-Fluo-4:** Ex 493 / Em 516 nm. Single wavelength. LOD: ~0.5 mM.
- Originally designed for intracellular Mg imaging; applicable to urine with calibration.

Raman Detection Mg^{2+} (monatomic ion) has no Raman-active modes. Not detectable.

FTIR Detection Mg^{2+} has no IR absorption. Not detectable by FTIR.

Voltammetry Detection

- Mg^{2+} cannot be reduced in aqueous media (E0 = -2.37 V vs SHE).
- **Potentiometric ISE** with Mg-selective ionophore is the main electrochemical approach.
- **Anodic stripping voltammetry** not applicable (too negative reduction potential).

Other Detection Technologies

1. **ICP-MS:** LOD ~0.00001 mg/dL. Multi-element, ultimate sensitivity.
2. **Flame emission photometry:** Mg emission at 285 nm. Less common than AAS.
3. **Paper-based colorimetric:** Calmagite-impregnated strips. LOD: ~0.5 mmol/L.

Summary Table: Detection Methods Comparison

Method	LOD	Key Parameter	Sample Prep	Strengths
Colorimetric (calmagite)	~0.04 mmol/L	520-550 nm	None	Standard, automated
Flame AAS	~0.001 mg/dL	285.2 nm	Dilution	Reference, specific
ICP-OES	~0.0001 mg/dL	279.6 nm	Dilution	Multi-element
Mag-Fura-2 fluorescence	~0.1 mM	Ex 340-380/Em 510 nm	Probe addition	Ratiometric
ISE	~0.1 mmol/L	Potentiometric	None	Real-time
Raman / FTIR	N/A	N/A	N/A	Not applicable

References

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